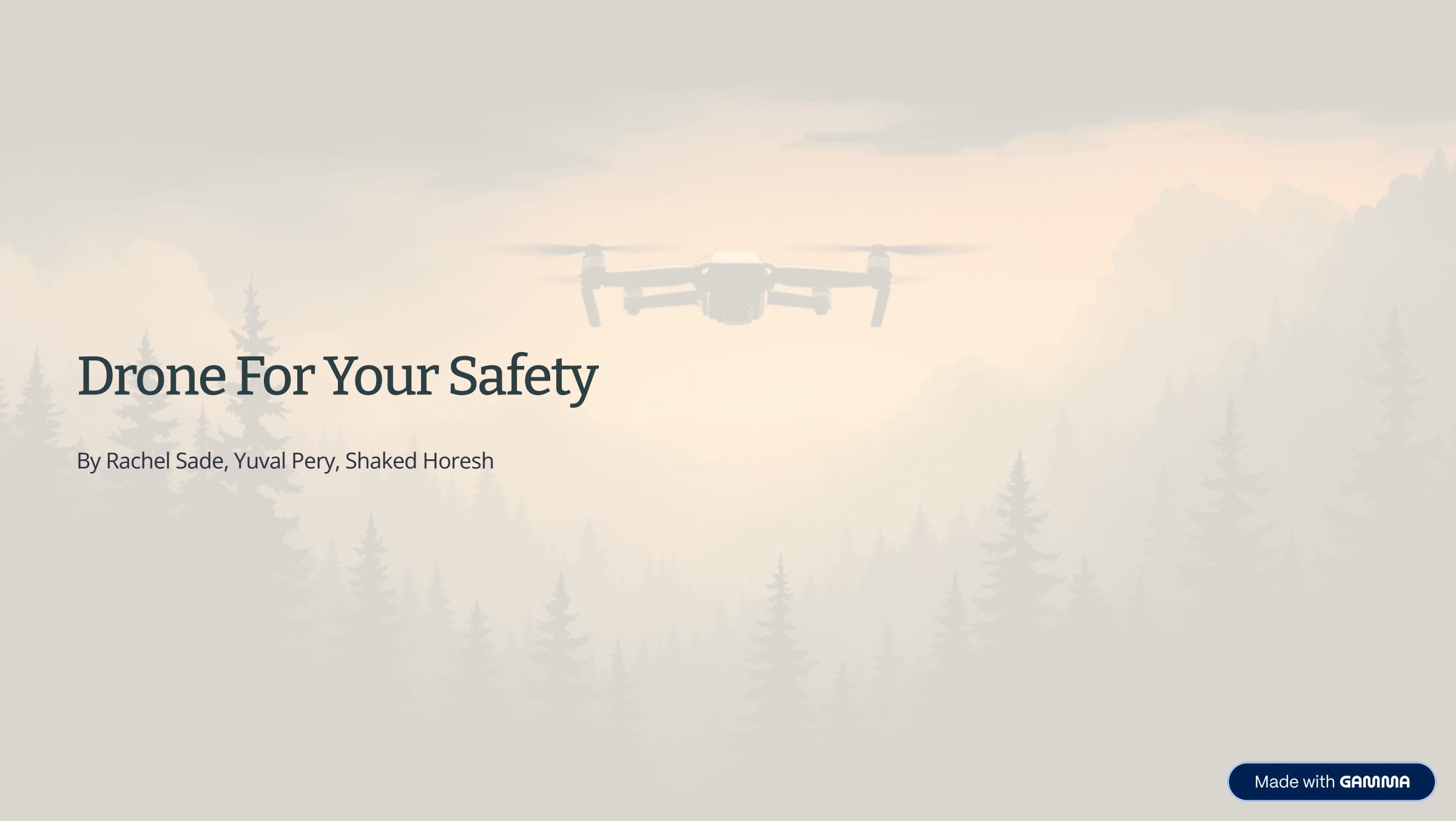


A drone is shown in flight, centered in the upper half of the frame. It is a quadcopter with a white body and black arms. The background is a soft-focus landscape of a forest with many evergreen trees. The sky is a warm, hazy orange, suggesting a sunset or sunrise. The overall tone is calm and scenic.

Drone For Your Safety

By Rachel Sade, Yuval Pery, Shaked Horesh

Adverse-Weather Drone Object Detection

In fires or smoke, visibility decreases - rapid identification is needed to support rescue and warning in order to save lives.



Task

Object Detection (Bounding Boxes)



Metrics

Precision, Recall, mAP



Models

YOLOv8 and YOLOv11

Key Achievements & Novelty

Project Achievements

- YOLO dataset: train/ val/ test
- Labels: person, fire, smoke
- Creating extreme conditions: fire + smoke on forest images
- YOLOv8 training + best.pt + evaluation on test
- Results: precision / recall / mAP + recognition examples



Novelty & Innovation

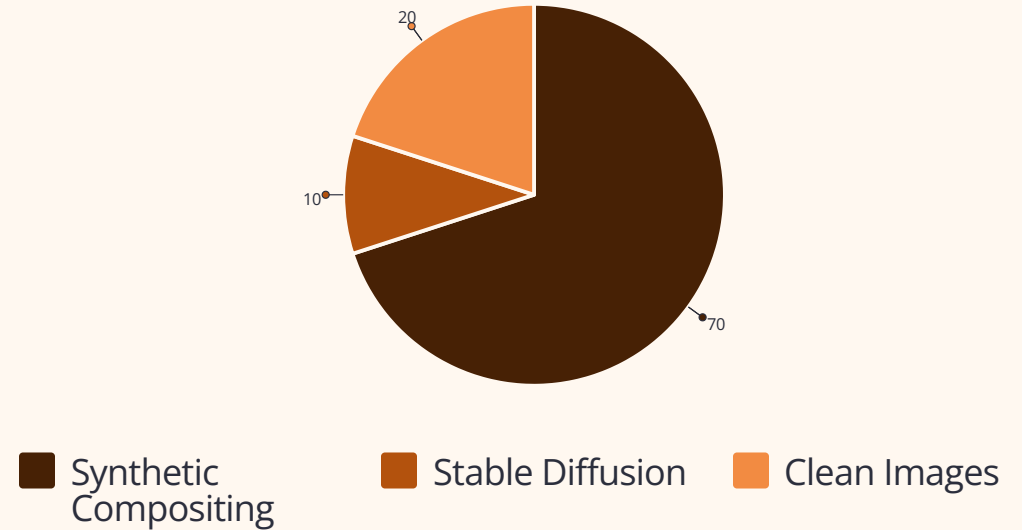
- Taking forest images and creating fire / smoke for an existing data collection
- Using the model even in low-quality image conditions
- Generating the fire and smoke to cover the people in the dataset



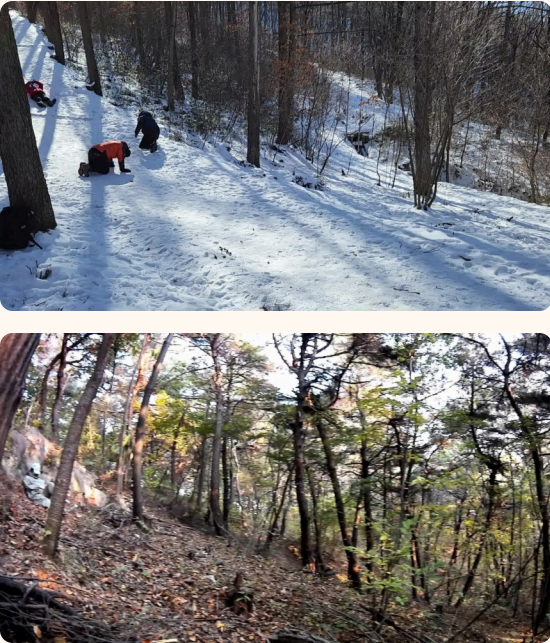
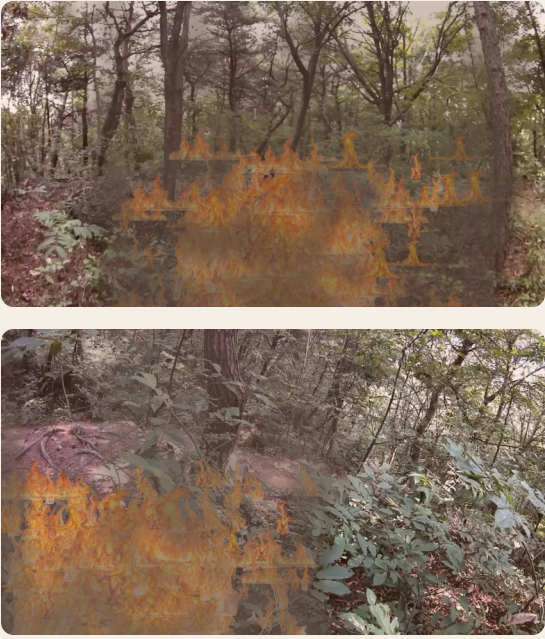
Dataset Composition by Generation Method

Why we used all three

- **Synthetic Compositing (70%)** – Provides precise ground truth and full control over labels for synthetic compositing.
- **Stable Diffusion version 1.5 (10%)** – Improves realism, especially for complex smoke / fire appearance and lighting consistency.
- **Clean (20%)** – Reduces false positives and trains the model on "no-fire / no-smoke" scenarios for robust detection.



Dataset Samples: Before vs After (Total = 2,500)

	Augmented Output	Source Image (Clean)	Sample Count
Original forest images (without fire)	None		500
Fire and smoke generated with Stable Diffusion			250
Fire and smoke generated with Synthetic Compositing			1750

Development Pipeline & Evaluation

1

Data Collection

Forest images + creating versions with fire/smoke

2

Labels

person/fire/smoke in YOLO format

3

Division

Train/ val/ test

4

Model Training

Fine-tuning → pretrained YOLOv8 and YOLOv11

5

Effort

We ran the image creation several times and tried different prompts until we got fire and smoke that looked real enough to train YOLO on.

Human Detection Performance:

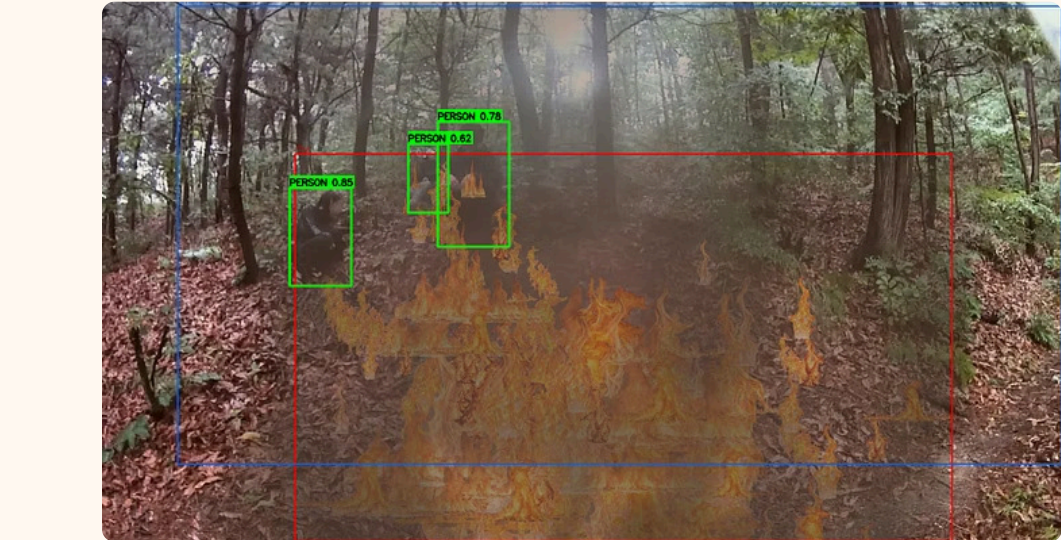
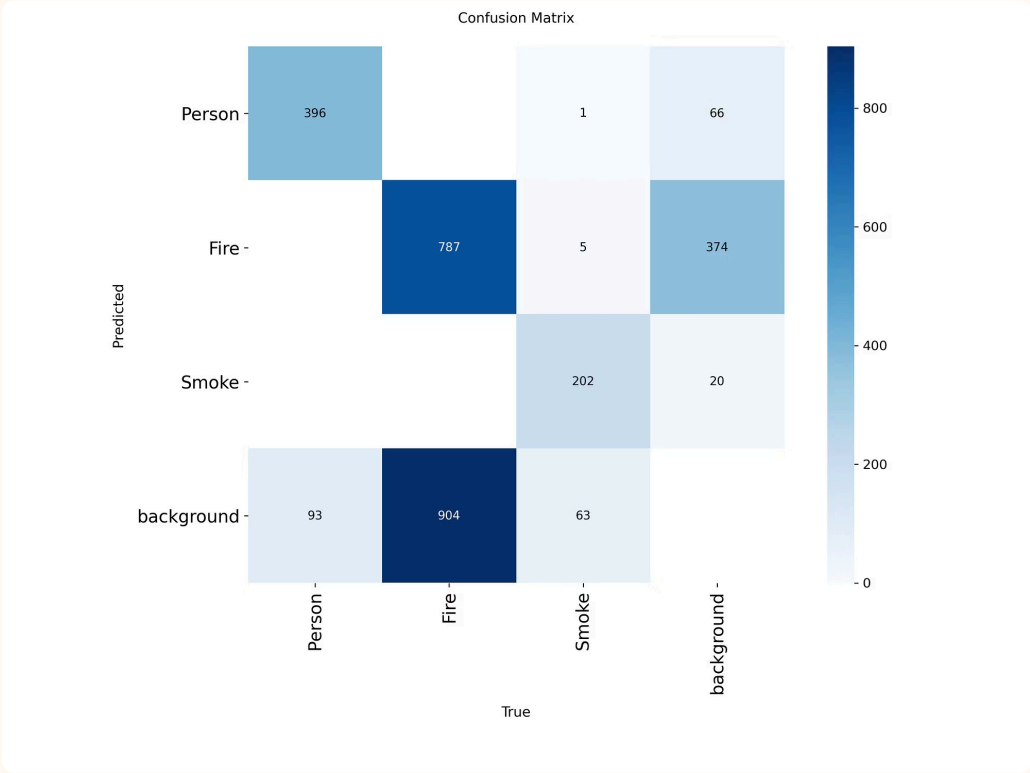
Person Detection Accuracy (Comparison: YOLOv8 vs. YOLOv11)

YOLOv8:

Precision = 69%
Measures the accuracy of positive predictions.

Recall = 70.9%
Indicates the model's ability to find all relevant instances.

mAP50 = 71.9%
Mean Average Precision at IoU 0.5, measuring bounding box accuracy.

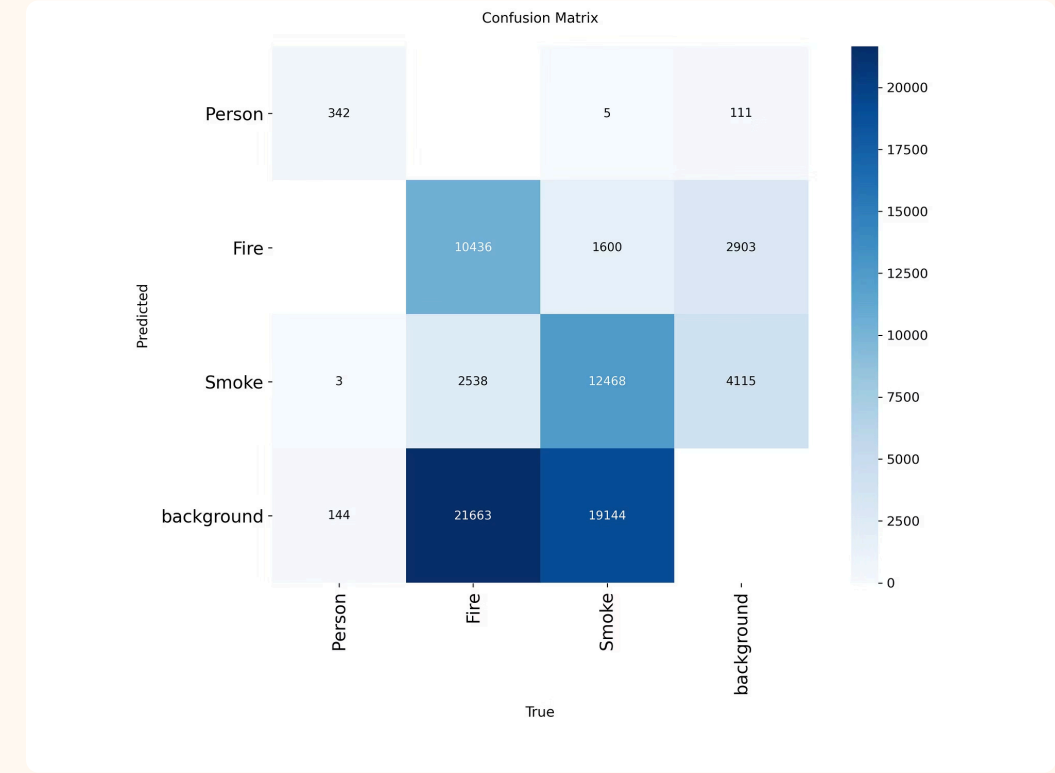


YOLOv11:

Precision = 84.4%

Recall = 79.6%

mAP50 = 82%



Project Outlook: Achievements, Learnings & Next Steps

We were able to train YOLOv8 & YOLOv11 to recognize people and fire

Key Takeaways & Learnings

- We learned that data is the most important factor, we need more variety
- In smoky conditions, objects simply disappear into the background, so we need data that simulates this so the model can learn to cope.

Future Work & Next Steps

- Expand and clean the data, then train again and compare improvement against the baseline
- Add real wildfire images for more realistic performance
- Extend beyond forests and fires to additional extreme scenarios at different places to support more rescue missions

